Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/26/2020
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
Contact Email:	Ye.Zhang-1@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	May 2020 report: On August, 2019, A Postdoctoral Fellow was hired to work on this grant. However, due to personal reasons, the postdoctoral fellow had to resign after only two weeks. On November 1, 2019 a Research Associate (Master's level) was hired to fill the position of the postdoctoral fellow who departed.		
COI Name (Institution):	Nakashima, Jin Ph.D. (Noble Research Institute)		
Grant/Contract No.:	80NSSC18K1462		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project. Innovative technologies developed through this grant include lines of new model plants such as barrel medic (Medicago truncatula) and purple false brome (Brachypodium distachyon) expressing genetically-encoded sensors to the cytoskeleton and cytoplasmic calcium. We expect that these new plant lines will significantly advance basic plant biology research on Earth.
Task Progress:	For this reporting period, we made significant progress with all three aims. An important milestone during this reporting period is the completion of aim 2. For this aim, we used a microgravity simulator called a clinostat to test the hypothesis that a natural plant compound called brassinolide regulates the direction of root growth through its effects on a cellular structure called the cytoskeleton (or cell skeleton). The two major components of the cytoskeleton are microtubules and microfilaments. The latter component is made up of a protein called actin; actins are proteins also found in muscle and nerve cells. From previous NASA-funded research, it has often been hypothesized that actin is major regulator of gravitropism (i.e., the process of shoots growing up and roots growing down). However, results from this aim, which we published this year, suggest that the role of actin has to do more with regulating autotropism or the straightening of plant organs after completing gravitropism. Our task progress during this reporting period also included the use of the Airbus Random Positioning Machine (RPM) and the Space Bio-Laboratories GRAVITE 3-D Clinostat at the Microgravity Simulation Support Facility (MSSF) located at the Kennedy Space Center (KSC). We used the RPM and 3-D clinostat to determine if changes in the composition, abundance, and distribution of cell wall components of plant roots were affected by simulated microgravity (aim 1). The cell wall of plants consists of a carbohydrate-rich network and some proteins embedded within this network. The cell wall is largely responsible for giving plants their rigidity, which allows their shoots to grow upright or roots to penetrate the soil. Our experiments at the MSSF have revealed that simulated microgravity can indeed modify certain cell wall properties in roots. However, the type of changes can differ depending on the type of microgravity simulator used. We also compared cell wall changes triggered by simulated microgravity versus those induced by spaceflight. The latter results were obtained from past NASA spaceflight projects. We found that the RPM mirrored more closely cell wall modifications induced by spaceflight. Plans are underway to repeat microgravity simulator experiments at the MSSL for at least two more times. For this reporting period, we also made significant progress with aim 3. For this aim, we evaluated root growth orientation in the model plant, Arabidopsis thaliana, that was disrupted in genes involved in actin function (i.e., actin mutants). Consistent with our results from aim 2, we found that actin mutants would reorient their root growth on a 2-D clinostat more strongly when compared to plants without any actin genes missing. These results further reinforce our conclusions from aim 2 that the actin cytoskeleton is a major regulator of plant autotropism. Experiments that we conducted in aim 3 also identified a new component of the cell that participates in maintaining the integrity of actin networks. This protein called SPIRRIG is enriched in parts of the cell where actin- and cell wall-dependent growth is most active, most notably the tips of root hair cells. A peer-reviewed paper that reports on the characterization of this protein is in preparation. Finally, we have introduced to the scientific community two Medicago truncatula (a model plant legume) lines that contain fluorescent reporters so their actin cytoskeleton can be studied in real time. These new F-actin reporter lines in

	M. truncatula could potentially be used to address a number of biological questions in legumes, particularly processes that do not occur in other model plants such as Arabidopsis. With regard to NASA relevance, a future gravitational biology-related study in which these new M. truncatula lines would be of value is the dynamic monitoring of F-actin in various tissues of graviresponding roots or with future light microscopy modules housed on the International Space Station (ISS).
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	Nakashima J, Yi M, Richard JT, Gilroy S, Blancaflor EB. "Immunocytochemistry of cell wall polysaccharides of Arabidopsis thaliana seedlings frown in microgravity analogs." Presented at the Investigator Poster Session, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Blancaflor EB. "Brassinolide inhibits autotropic root straightening through partial disruption of the actin cytoskeleton." Presented at the Concurrent Oral Session, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	De Bang L, Paez-Garcia A, Cannon AE, Chin S, Kolape J, Liao F, Sparks JA, Jiang Q, Blancaflor EB. "Brassinosteroids inhibit autotropic root straightening by modifying filamentous-actin organization and dynamics." Front Plant Sci. 2020 Feb 4;11:5. https://doi.org/10.3389/fpls.2020.00005 ; PMID: 32117357; PMCID: PMC7010715 , Feb-2020
Books/Book Chapters	Sadot E, Blancaflor EB. "The actomyosin system in plant cell division: Lessons learned from microscopy and pharmacology." in "The Cytoskeleton: Diverse Roles in a Plant's Life." Ed. V. Sahi, F. Baluška. Cham: Springer, 2019. p 85-100. (Plant Cell Monographs, vol. 24) https://doi.org/10.1007/978-3-030-33528-1_6 , Dec-2019
Significant Media Coverage	Chin S. "Research Update: Laying deeper roots on Earth and beyond. Article on research in the Blancaflor lab." Website Feature: Noble Foundation, April 22, 2020. https://www.noble.org/news/research-updates/laying-deeper-roots-on-earth-and-beyond/ , Apr-2020